

Saarth Vardhan

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Data Science M.S. at Columbia University with experience building forecasting, risk, and decision-support analytics on large, messy datasets across media, finance, and industry contexts.

Education

Columbia University in the City of New York MS in Data Science • Coursework: Probability & Statistics; Statistical Inference & Modeling; Machine Learning; Forecasting; Applications of Data Science in Insurance & Banking; Finance & Structuring for Data Science	New York, NY Expected Dec 2026
PES University B.Tech in Computer Science and Engineering (GPA: 3.92/4.0)	Bengaluru, India May 2025

Professional Experience

Bain & Company Analyst Intern, Private Equity (PEG) • Analyzed large, heterogeneous datasets across 15+ diligence engagements to build growth forecasts, demand projections, and downside risk scenarios in fintech and cloud sectors, supporting high-stakes investment decisions. • Applied statistical modeling and time-series analysis to translate messy financial and operational data into interpretable forecasts and scenario ranges, communicating uncertainty and key drivers to cross-functional teams.	Bengaluru, India Jan 2025 – Jun 2025
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Projects

Early Warning Risk Detection & Monitoring System Data Science Institute, Columbia University • Built a forecasting and early-warning analytics framework integrating macroeconomic, housing, banking, and credit-market indicators to identify emerging stress patterns in complex, interconnected systems. • Designed and evaluated time-series and machine-learning risk signals, explicitly analyzing false-positive/false-negative tradeoffs and signal stability to support real-world planning and decision-making. • Conducted robustness, sensitivity, and validation analyses across alternative feature sets and thresholds, producing interpretable outputs suitable for ongoing monitoring and operational use.	New York, NY Sep 2025 – Nov 2025
Multimodal & Agentic AI Systems for Decision Support Amazon Bedrock Innovation Challenge • Developed an end-to-end agentic AI decision-support system using Amazon Bedrock to synthesize insights from structured datasets and unstructured documents. • Implemented retrieval-augmented generation (RAG) pipelines, tool-use orchestration, and multi-step reasoning workflows to accelerate analysis while maintaining transparency and human-in-the-loop validation. • Built scalable, repeatable pipelines emphasizing clean Python, modular design, and reproducibility, demonstrating how GenAI-assisted workflows can support production-adjacent analytics.	New York, NY Sep 2025
Children's Rights and the Arms Industry Institute for the Study of Human Rights, Columbia University • Collected and integrated large, heterogeneous datasets from UN reports, NGO casualty databases, and humanitarian medical studies to quantify real-world impact under incomplete and noisy data conditions. • Performed cross-dataset validation and attribution analysis, linking weapons, manufacturers, and documented outcomes using structured trade data and qualitative reports, applying risk inference.	New York, NY Jan 2026 – Present

Skills

Programming & Data: Python, SQL, R, Pandas, NumPy, scikit-learn, data preparation, data validation
Modeling & Forecasting: Regression, time-series forecasting, clustering, anomaly detection, model evaluation
ML, GenAI & Analytics Workflows: Machine learning pipelines, AI-assisted analysis with validation, retrieval-augmented generation (RAG), agentic AI workflows, exposure to managed LLM services (Amazon Bedrock)
Systems, Visualization & Collaboration: Git (branching, pull requests), reproducible Python/SQL pipelines, QA and testing practices, Tableau, Power BI, Matplotlib, stakeholder-facing dashboards